

FINAL GROUP ASSIGNMENT — Data Preparation & Visualization Project

BANA 6760, Section HL1: Data Visualization

Term: Spring, 2026

Professor: Neba Nfonsang, PhD

Detailed Instructions

Group Size: 3–4 students

Tools Required: Tableau Desktop or Tableau Public, Python *or* Tableau Prep

Goal: Prepare data, join or stack datasets, aggregate datasets, build visualizations, create dashboards and a data story.

PROJECT OBJECTIVES

By the end of this assignment, your group will be able to:

- Locate and obtain **public datasets** that can be joined or stacked.
- Perform **data preparation** using either **Python** or **Tableau Prep**, applying techniques learned earlier in this course.
- Use Tableau to create **analytical visualizations** using best practices.
- Build **interactive dashboards** combining multiple charts.
- Create a Tableau **Story** to communicate insights clearly.
- Submit professional deliverables that demonstrate your end-to-end data analysis workflow.

DATA REQUIREMENT (Important)

Your group must locate **2 or 3 public datasets** that:

- Can be **joined** (e.g., shared keys such as CustomerID, ProductID, Date, Region, etc)

OR

- Can be **stacked/unioned** (same column structure)
- Choose datasets that are reasonably small (e.g., <200k rows) so Tableau performs well.

Examples of places to find free datasets:

- data.gov
- Kaggle
- Google Dataset Search
- UCI Machine Learning Repository
- UN Data
- IMF / World Bank Data
- Etc

PART 0: Analytic Questions and Goals

- State two or three questions you are trying to answer with your analysis.
- What is the overall goal of your analysis?
- Who are the consumers of your analysis results and dashboards/story?

PART 1: Data Preparation (Python or Tableau Prep)

Your team should apply **the skills** used in earlier assignments and this course using your own datasets.

You may choose **Python OR Tableau Prep** for this section.

You must perform **all the following**:

1. Data Cleaning

Your data cleaning should include tasks such as:

- Remove duplicates
- Identify and handle missing values
- Standardize text fields
- Fix inconsistent formats (dates, categories, numeric fields that contain text)

2. Data Structuring

Depending on your datasets:

- **Join** datasets using appropriate join keys
- **OR Stack** datasets with matching schemas
- Create aggregates
- Rename fields for clarity
- Ensure consistent field types across all datasets
- Produce a clean, final datasets such as **final_prepared_data** and **final_aggregate_data**

3. Data Enrichment

Create **at least 3 new calculated fields**, such as:

- Date-based parts (year, month, quarter)
- Column calculations
- Ratios or percentages
- KPIs
- Flags or categories based on conditions

4. Export the Clean Dataset

Save your output as:

✓ **final_prepared_data.csv**

You will use this file in Tableau.

PART 2 — Individual Worksheets Requirements in Tableau Desktop

Your group must create **six analytical worksheets**, each designed to highlight a different business question.

These should use best practices in:

- Chart selection
- Color and labeling
- Sorting and formatting
- Use of filters and interactivity
- Clean, readable presentation

1. Comparison Chart

Create a bar, column, or other comparison-based chart that compares a key metric across categories.

Examples you can choose from (pick based on your data):

- Revenue (or other metric) by category
- Count of items by group
- Total output by location
- Activity levels by demographic

Goal: Show which categories perform higher or lower.

2. Time-Based Trend

Create a line chart or time-series visualization that shows how a metric changes over time.

Examples:

- Monthly counts
- Daily totals
- Quarterly values
- Yearly comparisons
- (optional) Add a **linear trend** to the plot

Goal: Show trends, seasonality, and patterns over time.

3. Categorical Breakdown

Create a chart that shows the proportion of a key categorical variable.

Examples:

- Distribution of categories
- Breakdown of outcomes
- Classification results
- Types of events

This can be a:

- donut chart
- stacked bar
- tree map

Goal: Show how the data is split between categories.

4. Ranked Product/Item/Entity Analysis

Create a visualization that highlights top and bottom performers.

Examples:

- Top 5 best-performing items
- Lowest-rated categories
- Most frequent events
- Highest contributors to a metric

You may optionally:

- Create a calculated field to categorize items as “High” vs “Low” (or similar)
- Color code based on performance categories

Goal: Identify standout performers and underperformers.

5. Segment Analysis

Create a chart comparing **groups or segments** in your data.

Examples:

- Comparing demographic groups
- Comparing customer segments
- Comparing departments
- Comparing classes or categories

Use:

- bars
- stacked bars
- heatmaps
- or other segment-comparison charts

Goal: Compare how different segments behave or perform.

6. Bubble Plot

Create a **bubble plot** to show how **three variables** interact:

- **X-axis:** Numeric variable
- **Y-axis:** Numeric variable
- **Bubble size:** Third numeric variable (e.g., volume, quantity, count)
- **Bubble color:** Category or group (e.g., region, class, segment)

This visualization should reveal:

- patterns
- group differences
- clusters
- outliers

Include tooltips with additional context (e.g., names, categories, key metrics).

Name the worksheet: **Bubble Plot Insight**

7. Cross-Tabulation Table (Pivot Table)

Create a cross-tab table showing the sum of a numeric variable broken down by two categorical dimensions.

This could be something like:

- Sum of a metric by Category × Region
- Sum of Quantity by Product Type × Month
- Sum of Score by Department × Outcome
- Any two dimensions that make sense for your dataset

High-level requirements:

- Rows → one categorical field
- Columns → another categorical field
- Table values → SUM of a numeric metric
- Add color highlighting (optional) to emphasize high/low values

Name the worksheet: Cross-Tab Summary

PART 3 — DASHBOARD REQUIREMENTS

You should have at least **one dashboard**, each using multiple worksheets from Part 1.

8. Dashboard

Your group must create at least ONE polished dashboard that answers your analytic question and meets the needs of your intended stakeholders.

For example, a dashboard should include **at least three** of the charts you created in Part 2 such as:

- A comparison chart
- A time-trend chart
- A categorical breakdown chart
- a segment comparison chart
- a ranked performance chart
- a relationship/correlation chart

Also Include:

- at least two interactive filters
- a title and short insight summary
- a clean, tiled layout
- relevant filters (segment, category, time, etc.)
- interactive filtering and highlighting
- a short text box summarizing the key insights

Goal: Provide a broad understanding of your dataset's story.

PART 4: STORY REQUIREMENTS

9. Tableau Story: Multi-Step Narrative

Create a story that walks a user through your insights.

Your story must include at least **five** story points, such as:

1. Overview of your dataset
2. Comparison insights
3. Time-based trends
4. Segmentation or performance insights
5. Final recommendations

Each story point must include:

- a relevant visualization
- a caption summarizing what the viewer should notice

Goal: Communicate insights clearly and logically.

PART 5 — DELIVERABLES

Your group must submit:

✓ Packaged Tableau Workbook (.twbx)

Contains all sheets, dashboards, and storypoints.

✓ PDF exports

- Overview Dashboard

- Story Point PDF

✓ **Prepared Dataset(s) (CSV)**

Your final cleaned and enriched or prepared dataset generated from Python or Tableau Prep.

✓ **1-page Insight Summary (Word Document)**

Explains:

- The key findings from your analysis
- What the data suggests
- Business or practical recommendations
- Include a short citation or link for each dataset used in the comment section when you submit your work.